

Chapter 4.A

Chemical Reactions and Aqueous Solutions

Chemical Principles I

CHEM 1211K

1. Write complete and balanced chemical equations, based on either chemical symbols or word descriptions, including appropriate phases.
2. Recognize and describe five basic types of chemical reactions.
3. Describe and write equations representing the dissociation of ionic compounds upon dissolution in water.
4. Define and apply the terms strong electrolyte, weak electrolyte, and nonelectrolyte.
5. Identify compounds as being strong electrolytes, weak electrolytes, or nonelectrolytes from chemical formulas.
6. Apply solubility guidelines to predict the formation of a precipitate in reactions of ionic compounds in aqueous solution.
7. Write and interpret ionic and net ionic equations for precipitation reactions.

- **Chemical equations** represent the transformation of one or more chemical substances into new substances.
- **Reactants (starting materials)** are the initial components of a chemical reaction.
- **Products** are the components at the end of the reaction.
- Chemical equations are always written...

- A wealth of information can be extracted from chemical equations.
 - Stoichiometric coefficients** describe the numbers of each type of molecule or formula unit consumed or produced during the process.
 - Phase designators** indicate the state of matter of each substance at the macroscopic scale.

	Notation	Example
What happens?	Chemical formulas of the reactants and products	$\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2$
In what proportions?	Coefficients before each chemical formula	$2 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{H}_2\text{O}$
In what physical states?	(s), (l), (g) after each chemical formula	$2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{heat}} \text{H}_2\text{O}(\text{l})$
Under what conditions?	Special conditions written above or below the arrow	

- In a **balanced** chemical equation, the numbers of atoms of each element on the reactant side are equal to the numbers of atoms of each element on the product side.

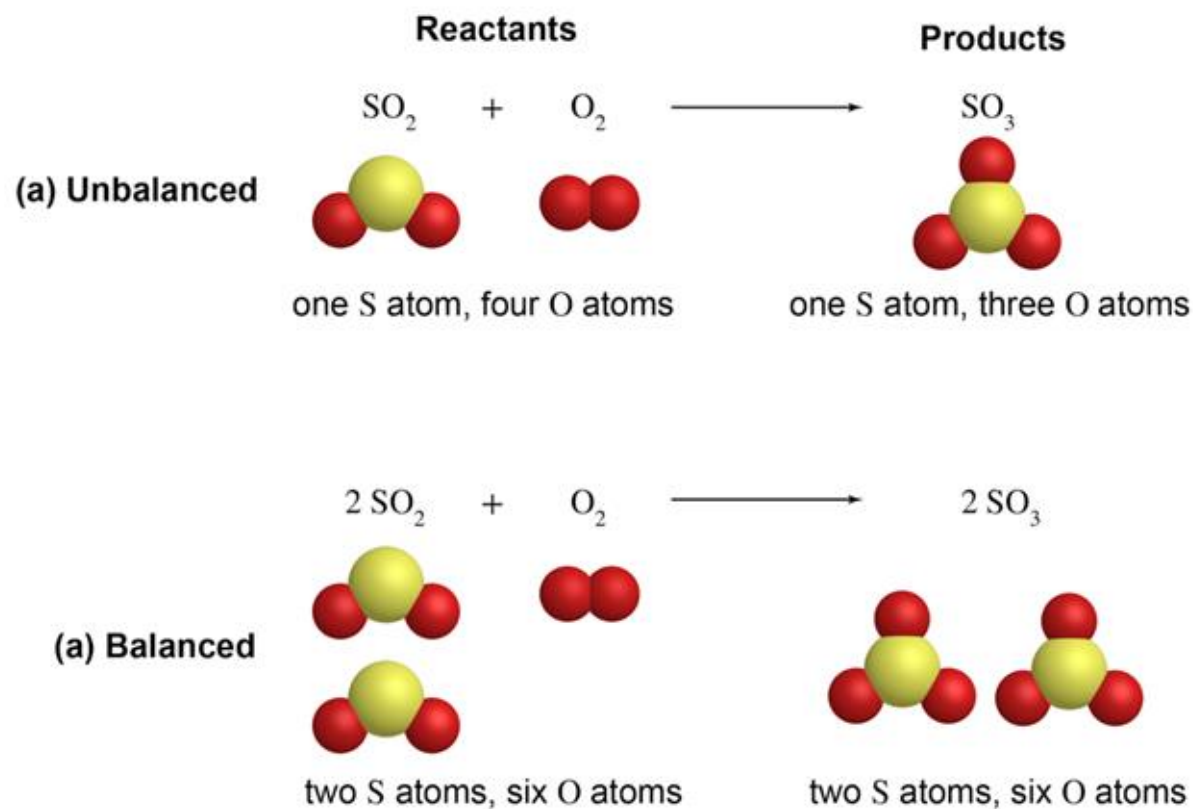


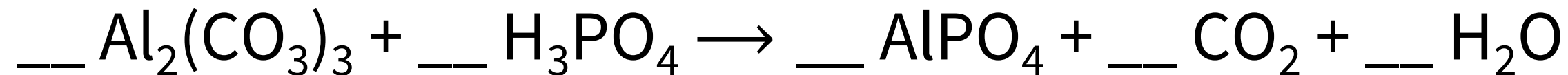
Figure 4.3. Unbalanced (a) and balanced (b) chemical equations.

- You should be able to rapidly balance chemical equations and easily verify that an equation is balanced.
- Stoichiometric coefficients are adjusted to balance chemical equations.
 - Do not modify subscripts in chemical formulas; that changes chemical identity. (Remember, compounds are defined by their formulas.)
 - Typically, we strive for the equation with the lowest whole-number coefficients possible, although fractional coefficients are completely fine.
- The process below can be used to balance an equation systematically.
 1. Balance unchanging polyatomic ions as units if they are present in both the reactants and products.
 2. Balance elements that appear in only one reactant and one product.
 3. Balance any remaining elements; resolve odd/even issues by doubling previously determined coefficients.
 4. Verify that the number of each type of atom is the same in the reactants and products and that the coefficients are in the simplest form.

Ammonia (NH_3) reacts with oxygen gas to form water and nitrogen dioxide.

Write chemical formulas for the reactants and products. What is the coefficient for oxygen when this reaction is balanced with the smallest possible whole-number coefficients? Enter your response as a numeral.

What is the coefficient for each chemical species when the following reaction is balanced with the smallest possible whole-number coefficients? To answer this question, enter the *sum* of all the coefficients.



- In this chapter, we'll explore five classes of chemical reactions.
 1. Synthesis reactions
 2. Decomposition reactions
 3. Single-replacement reactions
 4. Double-replacement reactions
 5. Combustion reactions

- **Synthesis reactions** involve the combination of simple reactants, often elements, to form a single more complex product.

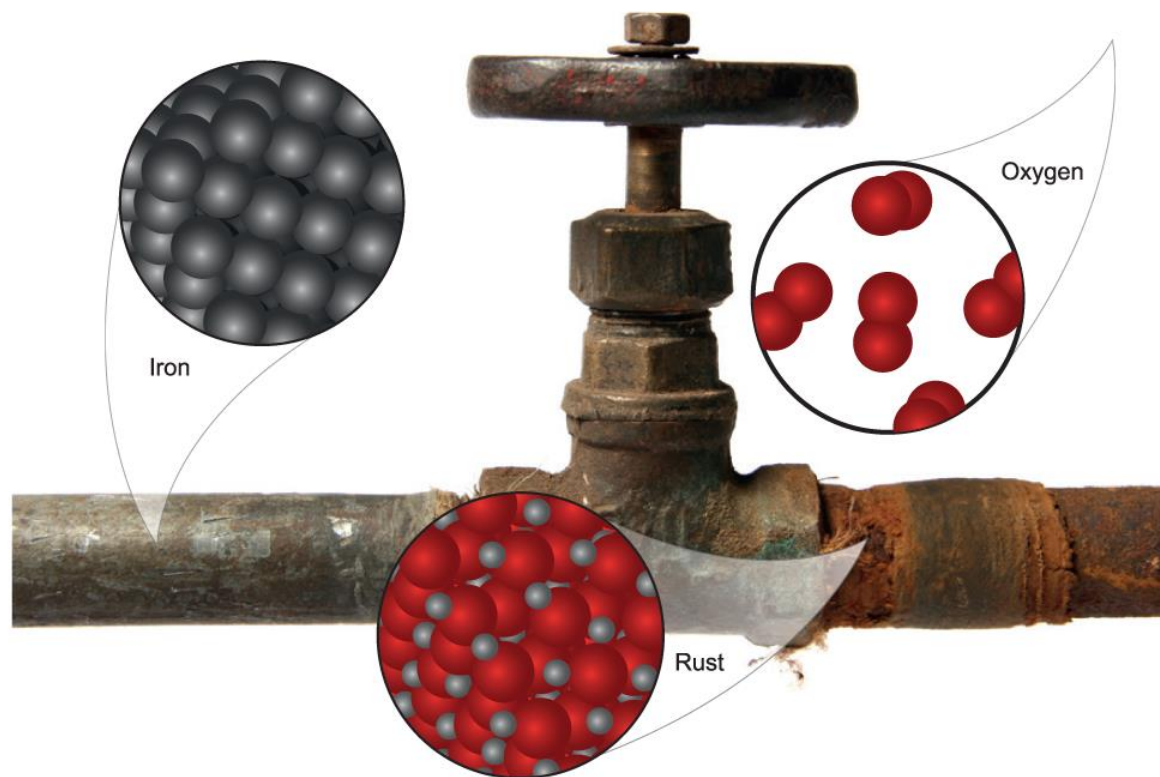
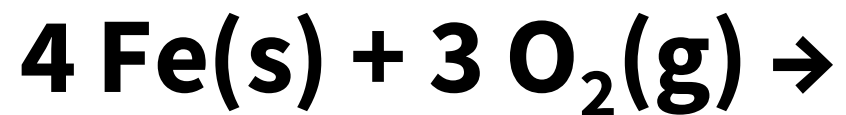


Figure 4.6. A synthesis reaction: formation of rust from iron and oxygen.

- **Decomposition reactions** involve the breakdown of a complex reactant into simpler products, often elements.

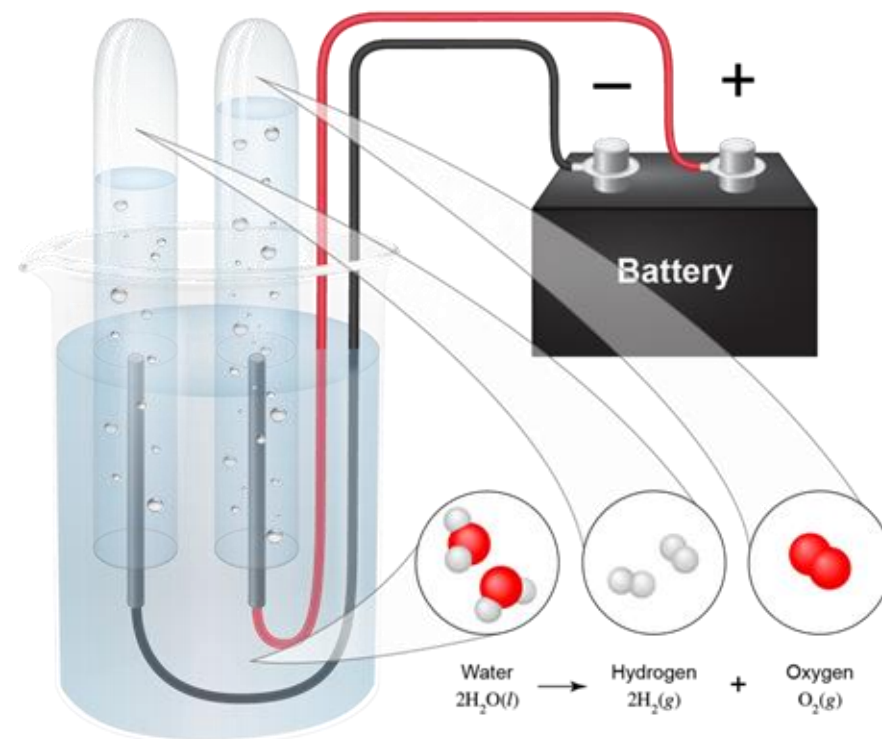
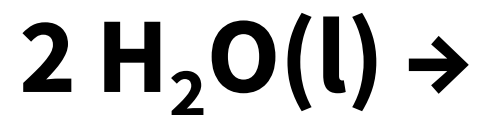


Figure 4.7. A decomposition reaction: electrolysis of water to hydrogen and oxygen gas.

- **Single-replacement (single-displacement) reactions** occur when an element reacts with a compound and displaces one of the elements in that compound, producing a new compound and a new element.

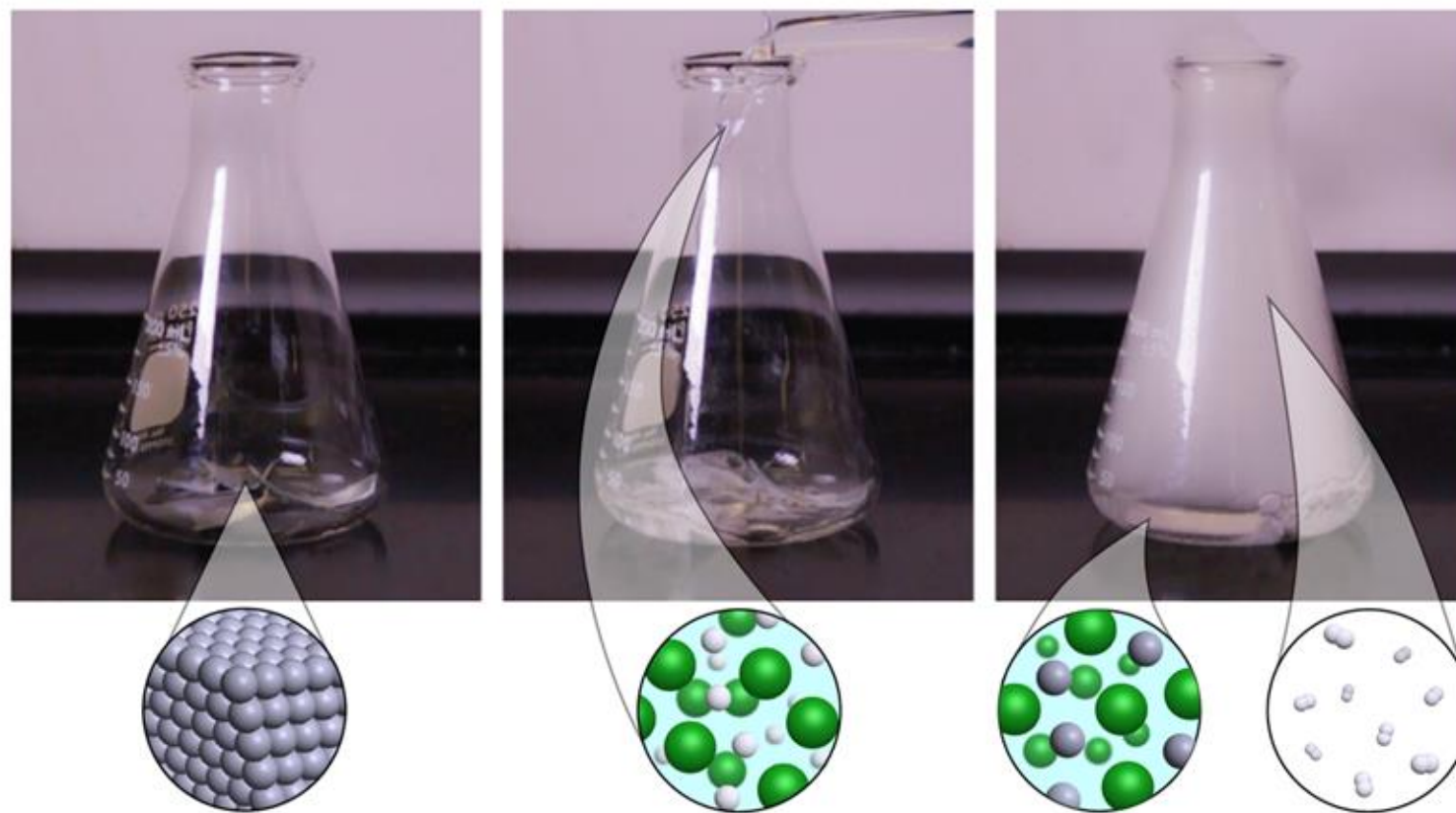


Figure 4.10. Single-replacement reaction of zinc with hydrochloric acid.

- **Double-replacement (double-displacement) reactions** occur when two aqueous ionic compounds exchange ions, forming two new compounds.
- In a **precipitation reaction**, one of the products is an ionic solid.

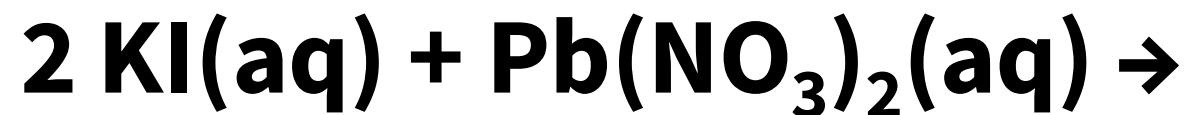
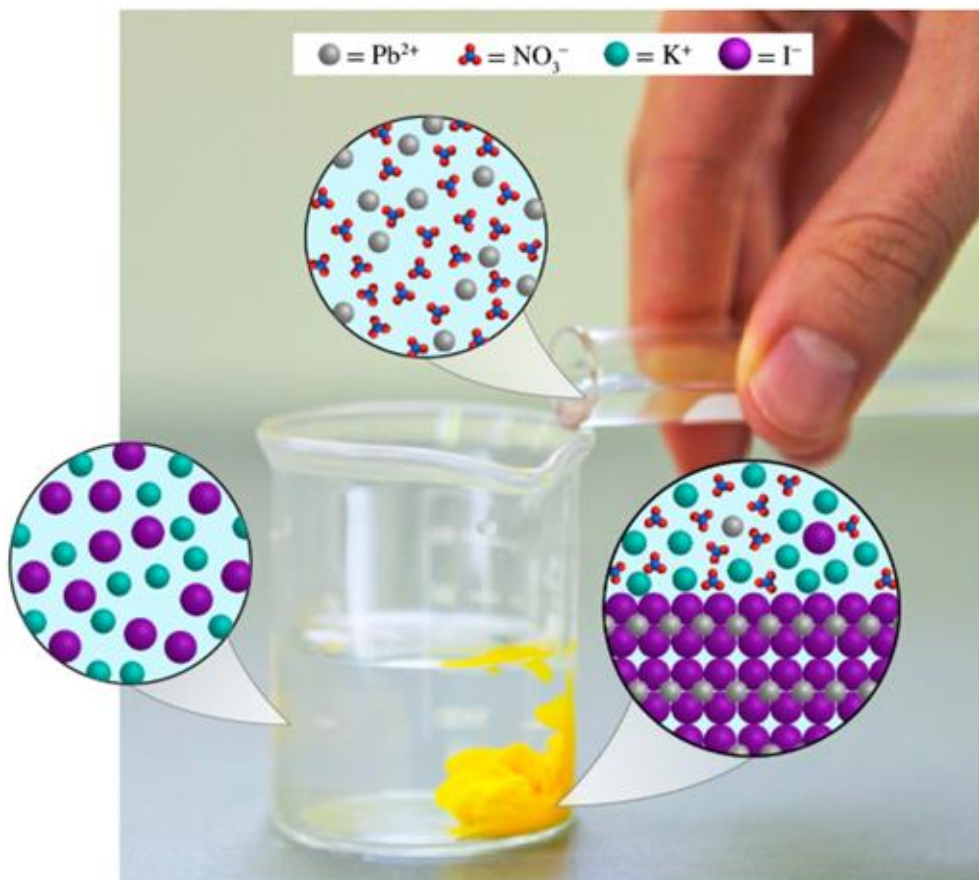
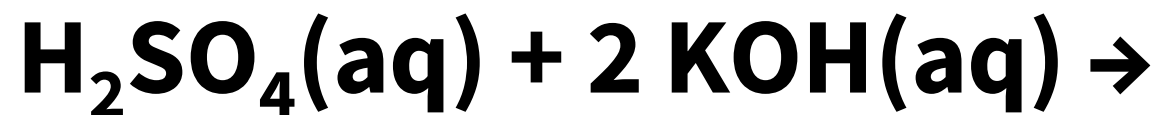
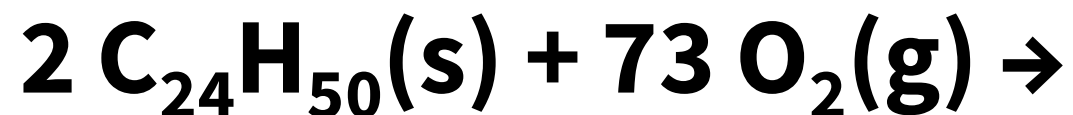
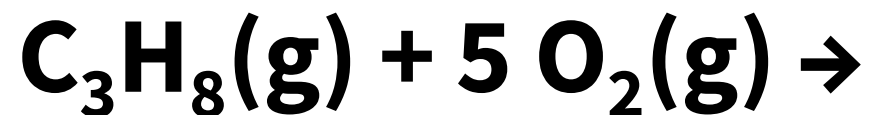


Figure 4.12. Double-replacement reaction of potassium iodide with lead(II) nitrate.

- **Acid-base reactions** are double-replacement reactions in which one of the “ions” exchanged is H^+ .
 - The **acid** is the reactant containing the H^+ transferred; the transferred H is often written first in the formula.
 - The **base** is the reactant that accepts or gains H^+ ; bases often contain the OH^- anion.



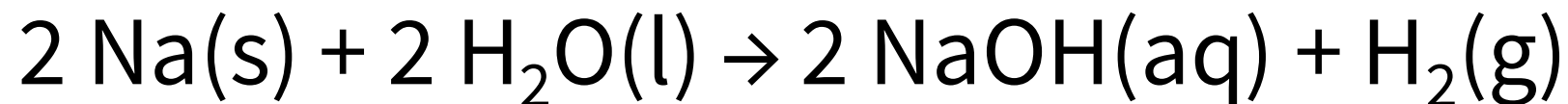
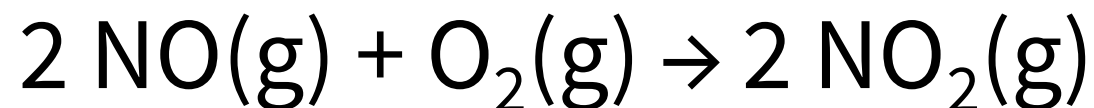
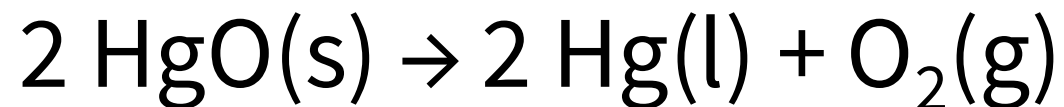
- **Combustion reactions** involve the rapid combination of a substance, often a hydrocarbon C_xH_y , with oxygen (O_2) to form carbon dioxide (CO_2) and water (H_2O).



- Summary of reaction types

Reaction Type	General Formula	Examples
Synthesis	$A + B \rightarrow AB$	$4 \text{ Fe(s)} + 3 \text{ O}_2\text{(g)} \rightarrow 2 \text{ Fe}_2\text{O}_3\text{(s)}$
Decomposition	$AB \rightarrow A + B$	$2 \text{ H}_2\text{O(l)} \rightarrow 2 \text{ H}_2\text{(g)} + \text{O}_2\text{(g)}$ $2 \text{ KClO}_3\text{(s)} \rightarrow 2 \text{ KCl(s)} + 3 \text{ O}_2\text{(g)}$
Single replacement	$A + BC \rightarrow AC + B$	$\text{Zn(s)} + 2 \text{ HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
Double replacement	$AB + CD \rightarrow AD + CB$	$2 \text{ KI(aq)} + \text{Pb(NO}_3)_2\text{(aq)} \rightarrow \text{PbI}_2\text{(s)} + 2 \text{ KNO}_3\text{(aq)}$ $\text{HBr(aq)} + \text{KOH(aq)} \rightarrow \text{KBr(aq)} + \text{H}_2\text{O(l)}$
Combustion	$\text{C}_x\text{H}_y + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$	$\text{C}_3\text{H}_8\text{(g)} + 5 \text{ O}_2\text{(g)} \rightarrow 3 \text{ CO}_2\text{(g)} + 4 \text{ H}_2\text{O(l)}$

Label each chemical reaction as *synthesis* (1), *decomposition* (2), or *single replacement* (3). To answer this question, enter a string of numbers labeling the reactions from top to bottom (e.g., “3121”).



Chapter 4.B

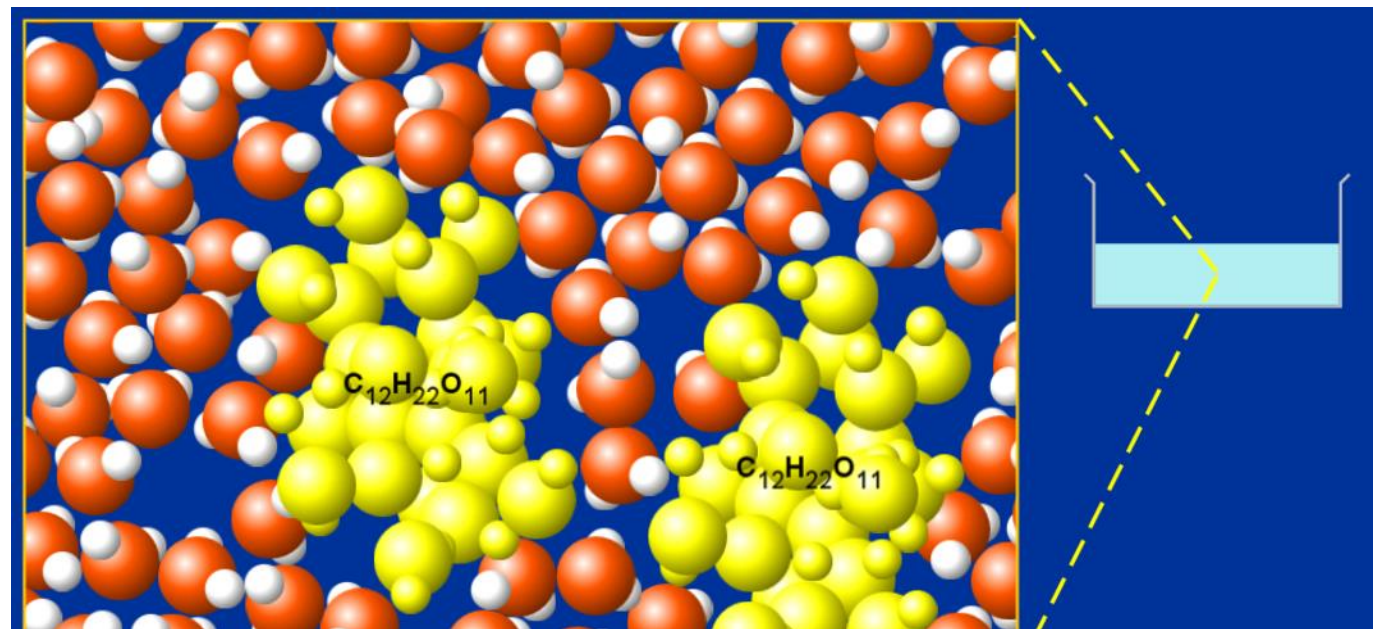
Chemical Reactions and Aqueous Solutions

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- **Aqueous solution**
- Solutions contain two types of components:
 - **Solvent**
- **Solute(s)**



- In all solutions, we say that *the solutes dissolve in (or are dissolved in) the solvent.*

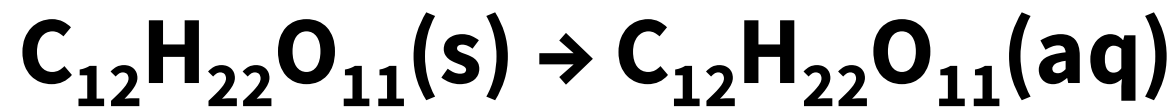
Rubbing alcohol is a solution that contains approximately 180 grams of isopropyl alcohol ($\text{C}_3\text{H}_8\text{O}$) for every 100 grams of water (H_2O). Identify the solute and solvent in this solution. How do you know?

Hint: masses can be misleading. What is the *number* ratio of $\text{C}_3\text{H}_8\text{O}$ to H_2O in this solution?

The solute is...

- A. Water
- B. Isopropyl alcohol

- The process of a compound dissolving into a solvent is called **dissolution**. Solute molecules are surrounded by solvent molecules and separate from one another. The solute “disappears into” the solvent.
 - Unreactive **covalent compounds** such as alcohols and sugars dissolve in water without any further change. For example, dissolution of the solid sugar sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in water is written as follows.



- When an **ionic compound** dissolves in water, the compound dissociates as each ion is surrounded by water molecules. This process is called **dissociation**; we generally assume it to be complete.

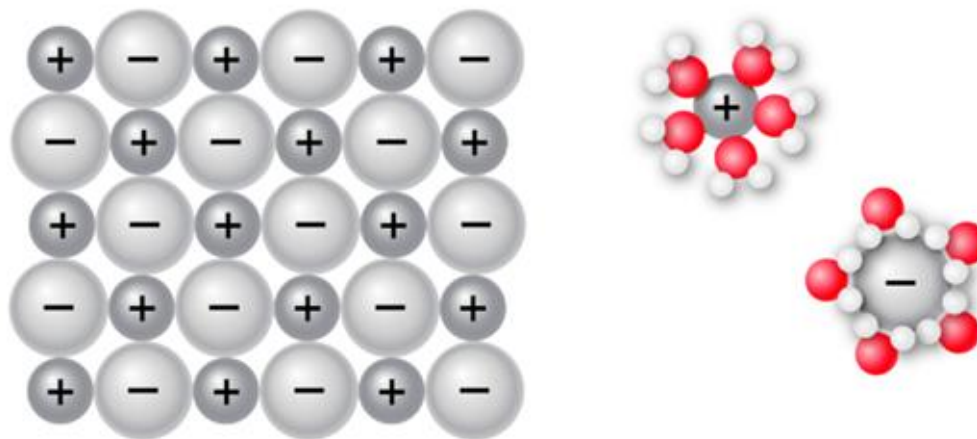
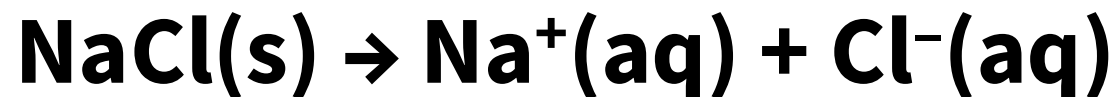


Figure 4.14. Sodium (+) and chloride (–) ions are surrounded by water molecules when sodium chloride is dissolved in water. The surrounded ions are said to be **hydrated**.

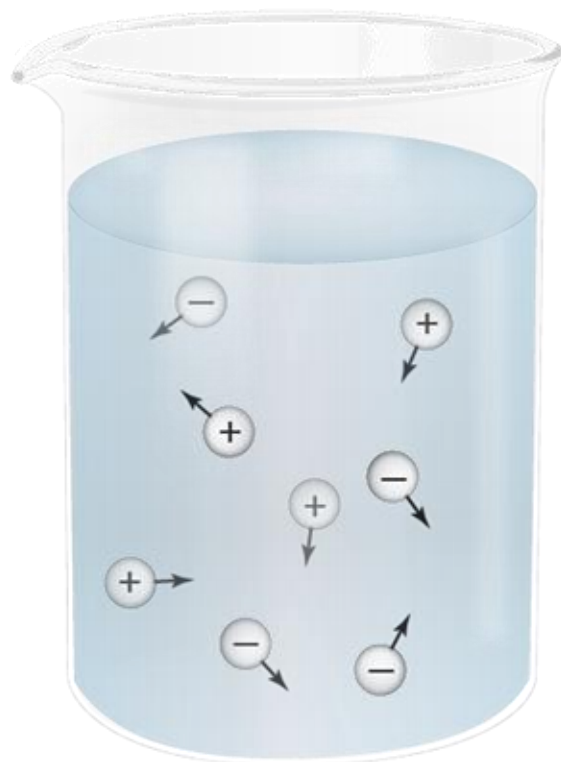
Which statement is *true* regarding dissociation and dissolution?

- A. Dissociation can occur without dissolution.
- B. Dissolution can occur without dissociation.
- C. Dissolution and dissociation can only occur together.
- D. None of these statements is true.

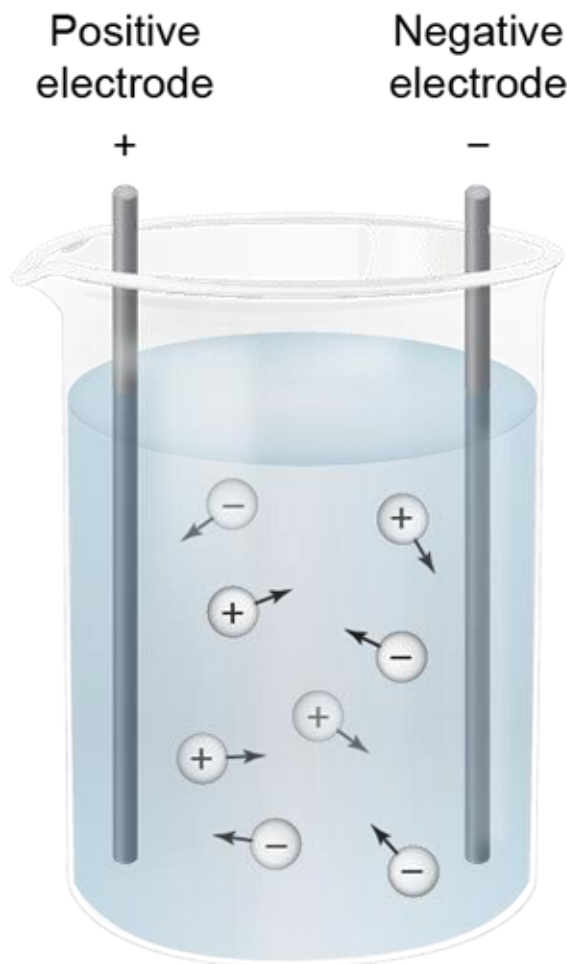
- We are often interested in whether a given can dissolve in a given solvent.
 - **Soluble**
 - **Insoluble**
- In CHEM 1212K, we will learn and apply the idea that solubility is a continuum. Throughout CHEM 1211K, we will treat solubility as a binary property. (It often is, essentially.)
- The **solubility rules** are a set of guidelines used to determine whether a given ionic salt (with known cation and anion) is soluble or insoluble in water.

- A solution that contains dissolved ions conducts electricity via motion of these ions; such solutions are called **electrolytes**.
- **Strong electrolyte**
- **Weak electrolyte**
- **Nonelectrolyte**

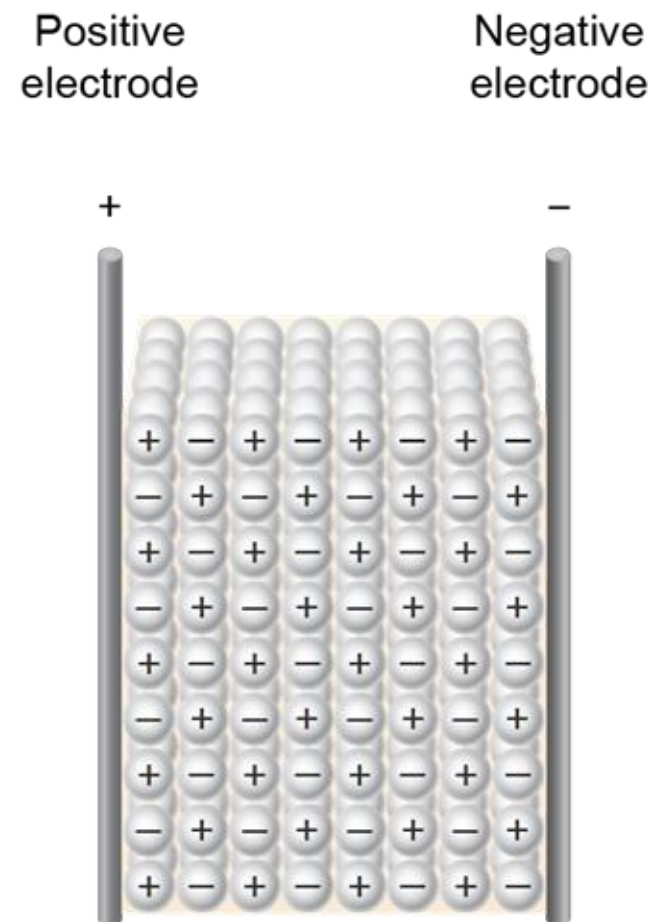
Compounds must be both soluble and ionic to be written in the form of their separate ions.



(a) Solution of ions



(b) Effect of charged electrodes



(c) Ionic solid

Figure 4.15. Mobile ions in aqueous solution make these solutions electrically conductive.

- Strong electrolytes include aqueous solutions of...
 - Soluble ionic compounds
 - **Strong acids:** compounds HX that, when dissolved, react *completely* with water to form $\text{H}_3\text{O}^+(\text{aq})$ and $\text{X}^-(\text{aq})$.
- Weak electrolytes include aqueous solutions of...
 - **Weak acids:** compounds HA that, when dissolved in water, react *partially* with water to form only small amounts of $\text{H}^+(\text{aq})$ and $\text{A}^-(\text{aq})$.
 - **Weak bases:** compounds B that, when dissolved in water, react *partially* with water to form only small amounts of $\text{OH}^-(\text{aq})$ and $\text{HB}^+(\text{aq})$.

- **Strong bases** are soluble hydroxide salts; these form strong electrolytes when dissolved in water. All other bases are weak, including insoluble hydroxide salts and amine bases.
- We will concern ourselves with seven (and *only* seven) **strong acids**. All other acids are weak...for our purposes.
- For H_2SO_4 , only the first proton transfer is complete. HSO_4^- is not a strong acid.
- HClO_3 is considered weak in some sources.

Compound	Formula	Ions
Lithium hydroxide	LiOH	Li^+, OH^-
Sodium hydroxide	NaOH	Na^+, OH^-
Potassium hydroxide	KOH	K^+, OH^-
Calcium hydroxide	$\text{Ca}(\text{OH})_2$	$\text{Ca}^{2+}, \text{OH}^-$
Strontium hydroxide	$\text{Sr}(\text{OH})_2$	$\text{Sr}^{2+}, \text{OH}^-$
Barium hydroxide	$\text{Ba}(\text{OH})_2$	$\text{Ba}^{2+}, \text{OH}^-$

Compound	Formula	Ions
Hydrochloric acid	HBr	H^+, Br^-
Hydrobromic acid	HCl	H^+, Cl^-
Hydroiodic acid	HI	H^+, I^-
Nitric acid	HNO_3	$\text{H}^+, \text{NO}_3^-$
Sulfuric acid	H_2SO_4	$\text{H}^+, \text{SO}_4^{2-}$
Perchloric acid	HClO_4	$\text{H}^+, \text{ClO}_4^-$
Chloric acid	HClO_3	$\text{H}^+, \text{ClO}_3^-$

- Figure 4.17 gives you an opportunity to explore different compounds to determine whether they are strong electrolytes, weak electrolytes, or nonelectrolytes.

Solution Type	Compound Type	Examples
Strong electrolyte	Soluble ionic salts	NaCl(aq), K ₂ SO ₄ (aq)
	Strong bases	NaOH(aq), KOH(aq)
	Strong acids	HCl(aq), HNO ₃ (aq)
Weak electrolyte	Weak acids	HNO ₂ (aq), H ₃ PO ₄ (aq)
	Weak bases	NH ₃ (aq), CH ₃ NH ₂ (aq)
Nonelectrolyte	Molecular and non-acidic	C ₆ H ₁₂ O ₆ (aq) and other sugars

Which of the following solutes produce an electrolyte solution (strong or weak) when dissolved in water?
Select all that apply.

- A. KOH
- B. sodium phosphate
- C. sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)
- D. copper(II) chloride
- E. hydrobromic acid
- F. HNO_2
- G. $\text{CH}_3\text{CH}_2\text{COOH}$
- H. HClO_4

Chapter 4.C

Chemical Reactions and Aqueous Solutions

Chemical Principles I

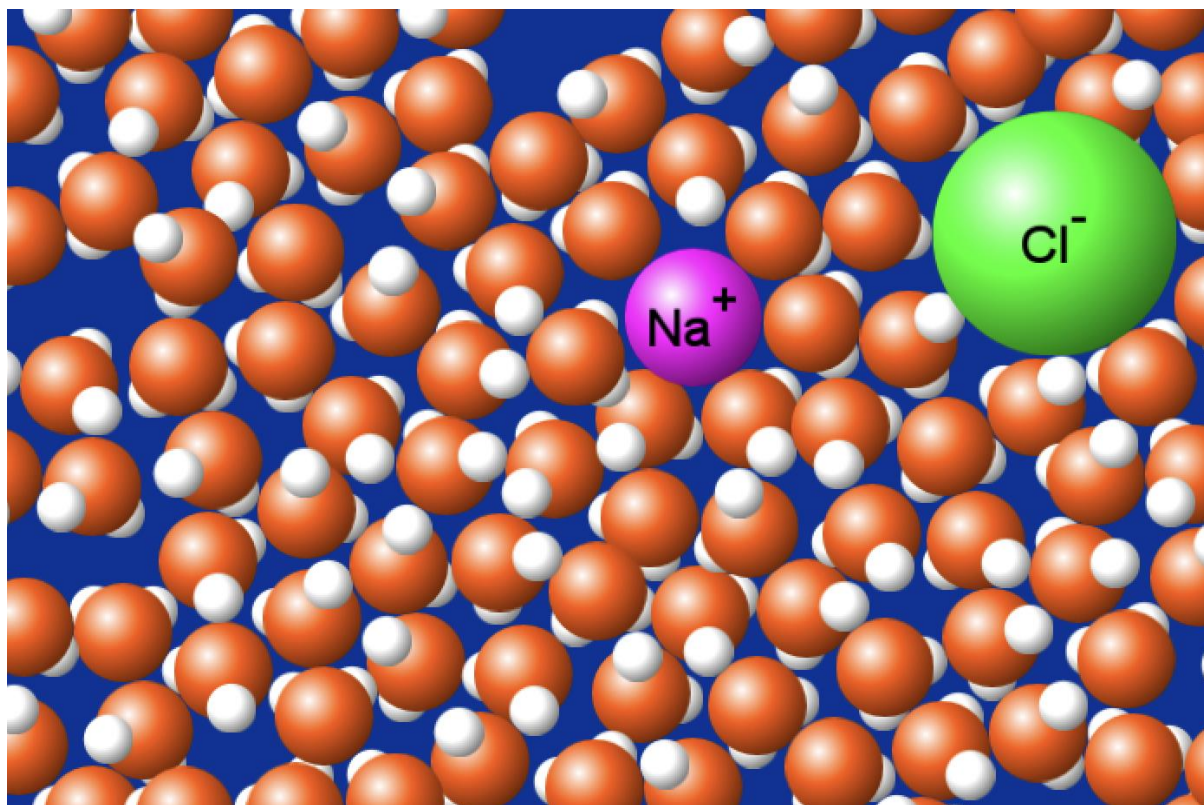
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7. Write and interpret ionic and net ionic equations for precipitation reactions.

Which of the following are considered weak acids? Select all that apply.

- A. Nitric acid
- B. HF
- C. CH_4
- D. Sodium hydroxide
- E. H_2SO_4
- F. Acetic acid

- Ionic compounds that readily dissolve in water are called **soluble**; those that do not readily dissolve in water are called **insoluble**.
- In CHEM 1212K, we will see that even insoluble compounds dissolve to an extremely tiny extent. For the purposes of CHEM 1211K (and many practical purposes) we can ignore this fact.



4.4 Precipitation Reactions | 2 of 4

- The **solubility rules** are a set of guidelines that enable us to efficiently determine whether a given ionic salt is soluble or insoluble. [Most guidelines include exceptions.](#)

Soluble ionic compounds contain these ions...	Exceptions
Group 1 cations: Li^+ , Na^+ , K^+ , Rb^+ , Cs^+ ; NH_4^+	—
$\text{C}_2\text{H}_3\text{O}_2^-$, HCO_3^- , NO_3^- , ClO_3^- , ClO_4^-	—
Cl^- , Br^- , I^-	Salts of Ag^+ , Hg_2^{2+} , and Pb^{2+}
F^-	Salts of group 2 cations, Pb^{2+} , and Fe^{3+}
SO_4^{2-}	Salts of Ag^+ , Ba^{2+} , Ca^{2+} , Hg_2^{2+} , Pb^{2+} , and Sr^{2+}

Insoluble ionic compounds contain these ions...	Exceptions
CO_3^{2-} , SO_3^{2-} , CrO_4^{2-} , PO_4^{3-}	Salts of group 1 cations and NH_4^+
S^{2-} , OH^-	Salts of group 1 cations and Ba^{2+}

Which statement is *true* regarding an aqueous solution of barium hydroxide?

- A. The solution contains many intact $\text{Ba}(\text{OH})_2$ formula units.
- B. The solution contains equal amounts of barium ions and hydroxide ions and very few (if any) $\text{Ba}(\text{OH})_2$ formula units.
- C. The solution contains equal amounts of barium ions, hydroxide ions, and $\text{Ba}(\text{OH})_2$ formula units.
- D. The solution contains equal amounts of barium ions and $\text{Ba}(\text{OH})_2$ formula units. It contains twice as many hydroxide ions as barium ions and $\text{Ba}(\text{OH})_2$ formula units.
- E. The solution contains twice as many hydroxide ions as barium ions, and it contains very few $\text{Ba}(\text{OH})_2$ formula units.

4.4 Precipitation Reactions | 3 of 4

- When two aqueous solutions of ionic compounds are mixed, ions may come into contact that can form an insoluble ionic salt. The result is a **precipitation reaction**.

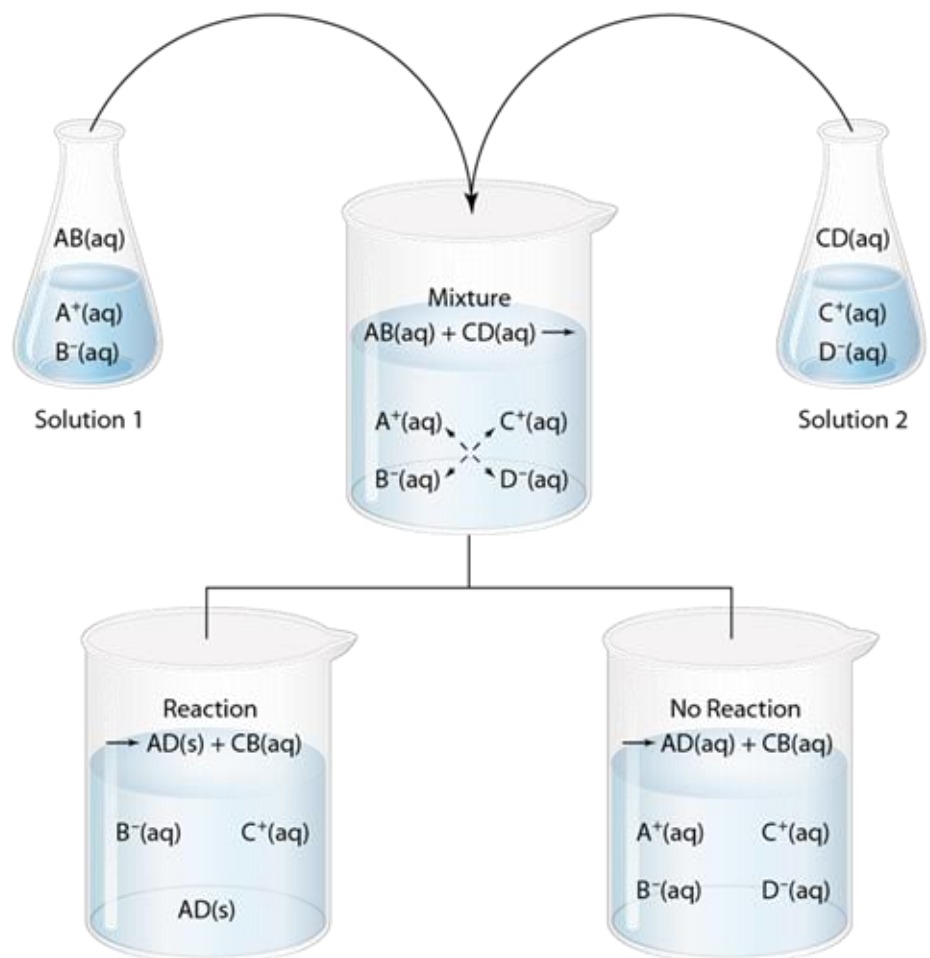


Figure 4.19. Mixing two aqueous solutions may result in formation of a **precipitate** AD (left) or no reaction (right).

- When considering two ionic salts in solution together, we “switch the dance partners” and look for an insoluble salt using the solubility rules. If none exists, no reaction occurs.
- The insoluble salt that forms is called a **precipitate**. The ions remaining in solution didn’t undergo any chemical change; we call them **spectator ions**.
- **Example.** Write a balanced chemical equation for the reaction of potassium sulfate with barium nitrate.

What is the *name* of the insoluble product formed when potassium carbonate reacts with iron(III) carbonate?

4.4 Precipitation Reactions



- A **total equation** or **molecular equation** lists complete chemical formulas for dissolved ionic compounds, without consideration of dissociation in aqueous solution.
- A **complete ionic equation** shows each *aqueous* ion as a separate species with its charge. Spectator ions appear on both the reactant and product sides.

Subtracting spectator ions from both sides gives the **net ionic equation**, which focuses on species undergoing chemical change.

Example. Write the total ionic equation for the reaction that occurs when aqueous solutions of aluminum sulfate and barium chloride are mixed.

Identify the number and type of spectator ions in the reaction of aqueous solutions of aluminum sulfate and barium chloride. Which of the following are spectator ions? After making this determination, write the net ionic equation for the reaction.



- Some of the amazing things we can now do with chemical reactions and aqueous solutions...
 1. Interpret chemical equations to infer the number ratios in which substances are consumed and produced, the states of matter of the reactants and products, and the conditions of the reaction.
 2. Balance chemical equations.
 3. Classify chemical reactions as synthesis, decomposition, single replacement, double replacement, or combustion processes.
 4. Describe and visualize solutions, particularly aqueous solutions of molecular (covalent) and ionic solutes; distinguish between dissolution and dissociation.
 5. Identify a given aqueous solution as a strong, weak, or non-electrolyte.
 6. Apply the solubility rules to infer whether a given ionic salt is soluble or insoluble in water.
 7. Predict the outcome with respect to precipitation (or not) when aqueous solutions of ionic salts are mixed with one another.
 8. Write molecular, total ionic, and net ionic equations for precipitation reactions.